

EXPERIMENT-3

1. Write a Java program that counts how many times a given number appears in an array.

Example:

Input: arr = [2, 3, 2, 5, 2, 6], target = 2

Output: 3

array.java	Output
<pre>1- public class array { 2- public static void main(String[] args) { 3- int[] arr = {2, 3, 2, 5, 2, 6}; 4- int target = 2; 5- int count = countOccurrences(arr, target); 6- System.out.println("The number " + target + " appears " + 7- count + " times in the array."); 8- } 9- public static int countOccurrences(int[] arr, int target) { 10- int count = 0; 11- for (int num : arr) { 12- if (num == target) { 13- count++; 14- } 15- } 16- return count; 17- } 18- }</pre>	The number 2 appears 3 times in the array. === Code Execution Successful ===

2. Given an array containing N-1 unique numbers from 1 to N, write a Java program to find the missing number.

Input: [1, 5, 6, 2, 4]

Output: 3

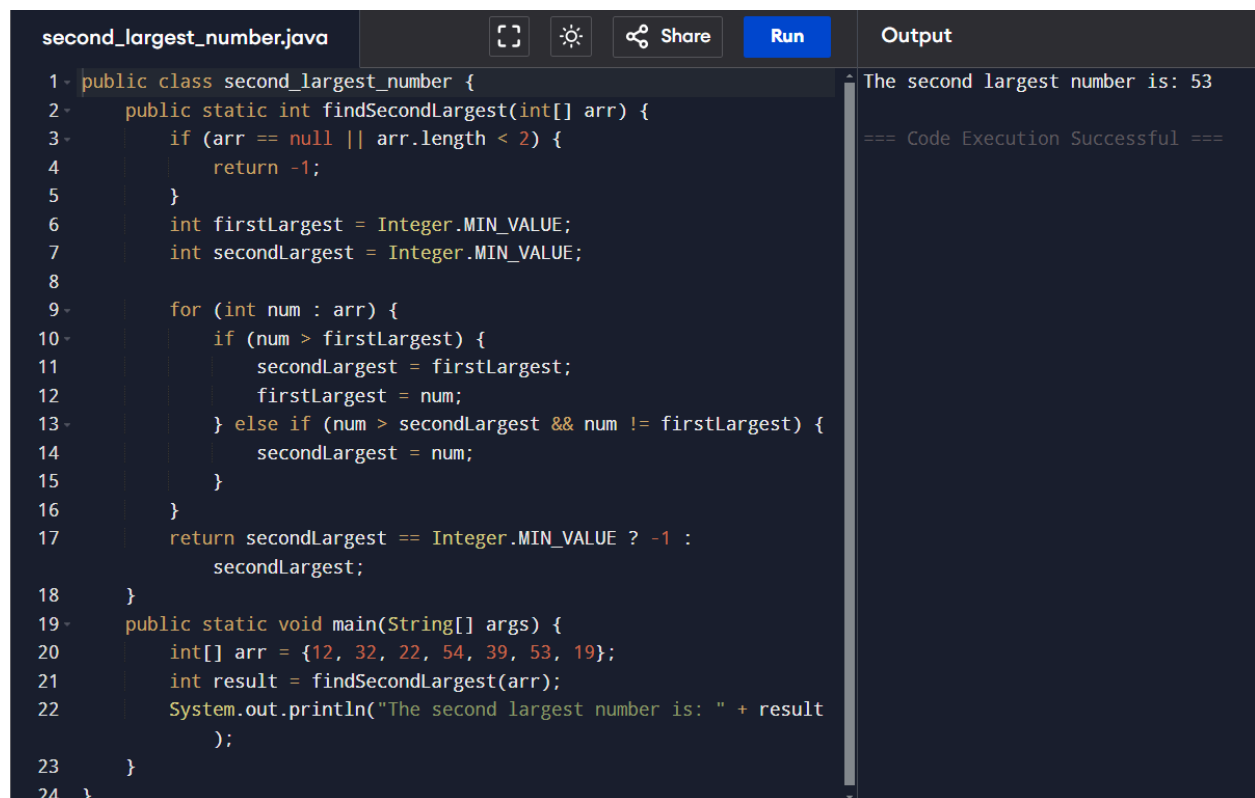
missing_number.java	Output
<pre>1- public class missing_number { 2- public static int findMissingNumber(int[] nums, int n) { 3- int totalSum = n * (n + 1) / 2; 4- int arraySum = 0; 5- for (int num : nums) { 6- arraySum += num; 7- } 8- return totalSum - arraySum; 9- } 10- 11- public static void main(String[] args) { 12- int[] nums = {1, 5, 6, 2, 4}; 13- int n = 6; // Since the array should contain numbers from 1 to 14- 6 15- int missingNumber = findMissingNumber(nums, n); 16- System.out.println("The missing number is: " + missingNumber); 17- } 18- }</pre>	The number 2 appears 3 times in the array. === Code Execution Successful ===

3. Write a Java program to find the second largest element in an integer array without sorting the array, if not exist return -1.

Example:

Input: arr = {12,32,22,54,39,53,19}

Output: 53

A screenshot of a Java IDE with a dark theme. The editor shows a file named 'second_largest_number.java'. The code defines a class 'second_largest_number' with a static method 'findSecondLargest' that takes an integer array 'arr' and returns the second largest element. It uses two variables, 'firstLargest' and 'secondLargest', initialized to 'Integer.MIN_VALUE'. A for loop iterates through the array, updating these variables. If the array is null or has fewer than 2 elements, it returns -1. The 'main' method tests the function with the array {12, 32, 22, 54, 39, 53, 19} and prints the result. The output pane on the right shows 'The second largest number is: 53' and '=== Code Execution Successful ==='.

```
second_largest_number.java
1 public class second_largest_number {
2     public static int findSecondLargest(int[] arr) {
3         if (arr == null || arr.length < 2) {
4             return -1;
5         }
6         int firstLargest = Integer.MIN_VALUE;
7         int secondLargest = Integer.MIN_VALUE;
8
9         for (int num : arr) {
10            if (num > firstLargest) {
11                secondLargest = firstLargest;
12                firstLargest = num;
13            } else if (num > secondLargest && num != firstLargest) {
14                secondLargest = num;
15            }
16        }
17        return secondLargest == Integer.MIN_VALUE ? -1 :
            secondLargest;
18    }
19    public static void main(String[] args) {
20        int[] arr = {12, 32, 22, 54, 39, 53, 19};
21        int result = findSecondLargest(arr);
22        System.out.println("The second largest number is: " + result
23        );
24    }
25 }
```

Output

The second largest number is: 53

=== Code Execution Successful ===

4. Given two strings s and t, return true if t is an anagram of s, and false otherwise.

Example 1:

Input: s = "anagram", t = "nagaram"

Output: true

Example 2:

Input: s = "rat", t = "car"

Output: false

string.java

Share

Run

```
1- import java.util.Arrays;
2- import java.util.Scanner;
3- public class string {
4-     public static boolean isAnagram(String s, String t) {
5-         if (s.length() != t.length()) {
6-             return false;
7-         }
8-         char[] sArray = s.toCharArray();
9-         char[] tArray = t.toCharArray();
10-         Arrays.sort(sArray);
11-         Arrays.sort(tArray);
12-         return Arrays.equals(sArray, tArray);
13-     }
14-     public static void main(String[] args) {
15-         Scanner scanner = new Scanner(System.in);
16-         System.out.println("Enter the first string:");
17-         String s = scanner.nextLine();
18-         System.out.println("Enter the second string:");
19-         String t = scanner.nextLine();
20-         scanner.close();
21-
22-         boolean result = isAnagram(s, t);
23-         System.out.println(result);
24-     }
25- }
```

Output

Enter the first string:
hello
Enter the second string:
hey
false

=== Code Execution Successful

Output

Enter the first string:
hey
Enter the second string:
hey
true